

Layout-Conditioned Flow Matching with Mask-Guided Region Pooling for Anime Face Generation: A Complete Record of Methods and Decisions

tinyflow project
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Abstract

We document, end to end, the construction of a 64×64 anime-face generator based on flow matching with x-prediction in pixel space, trained from scratch on 21,551 images with one RTX 4090. The record covers the data pipeline, the network to the level of every layer’s normalisation and initialisation, the objective, optimiser, sampler and evaluation protocol, and—at equal weight—every design decision, including the ones that failed. The central negative result is a derivation, confirmed by ablation, that no auxiliary loss on the predicted clean image against the target (edges, lines, palettes, mask-weighted variants, per-pixel re-weightings) can improve a flow-matching model: at the objective’s optimum such penalties are almost entirely irreducible and only bias the network toward hedged predictions. The central positive result is a structural change: conditioning on a three-channel semantic layout (face, eyes, mouth hulls from a landmark detector), sampled at test time from a Gaussian-mixture prior over a point-distribution model of the landmarks, together with *mask-guided region pooling*—a zero-initialised layer that pools decoder features inside each semantic region and broadcasts a projection back into that region only. It reduces the rate of mismatched iris colours from 35% to the data’s own 4–5% at a cost of 3.5 FID against a schedule-matched unconditioned control. We also report where the model fails (coverage: recall 0.15 against a real-vs-real ceiling of 0.77), the learning-rate instability that shaped the final training schedule, and three sampling-time methods that need no retraining: edge guidance (−2 to −8 FID), re-noise-and-re-solve refinement (−3), and *autoguidance* with a 40-epoch checkpoint of the same recipe as the guide (34.6 → 21.2; 33.2 → 18.1 on the best checkpoint), which takes the layout-conditioned model 11 FID past the unconditioned control. The recipe transfers to CelebAMask-HQ at 64×64 with real label maps as layouts (FID 27.3, 18.6 with autoguidance).

1 Introduction

This paper is a methods record rather than a claim of state of the art. Its purpose is to let a reader reproduce every number, understand why each component is what it is, and avoid repeating experiments that were tried and closed. It accompanies the `tinyflow` repository; where a symbol is defined in code, the file and function are named.

The starting point was a working pixel-space flow-matching model that shipped with two domain-motivated auxiliary losses (a Sobel gradient-difference loss and a dark-line “ink” loss), the intuition being that anime is line art over flat fills. The campaign asked

whether such dataset-specific inductive biases help. Answering that required (i) a theory of what an auxiliary loss can do at the flow-matching optimum (§5), (ii) an offline test-bed that does not need training (§6), (iii) paired ablations with a measured noise floor (§7), and (iv) a diagnosis of where the best model actually fails (§10). The diagnosis pointed at two distinct problems—a long-range consistency failure (mismatched iris colours) and a coverage failure—and the first of these is solved by the architecture of §12.

Contributions, in the order they were established:

1. A derivation (§5) that training-time losses on $\hat{x}$ against x_1 are irreducible at the conditional-mean optimum, and that x_1 -dependent per-pixel weights tilt the learned field; both confirmed by ablation (§7).
2. A measurement protocol (§4): FID noise, loss as the sensitive statistic for architecture changes, a validated iris-mismatch metric, and a 16-step sampler for evaluation.
3. A layout prior over detector landmarks (§3.6) and a conditioned model with mask-guided region pooling (§3.5), the first structural change with an unambiguous, reproduced win.
4. A bottleneck analysis (§10) locating the remaining gap in coverage and in small low-variance features (mouth, nose).

2 Data

Images. 21,551 anime face crops at $64 \times 64 \times 3$ (`data/animefaces.py`). PNGs are loaded once, converted to `float32`, mapped to $[-1, 1]$ by $x/127.5 - 1$ and cached (`.preprocessed/anime_faces.npy`). All public boundaries use (H, W, C) layout; only the U-Net’s interior is channels-first. Images are denormalised for display with `clip((x + 1) · 127.5, 0, 255)`; clipping is mandatory because the network has no bounded output activation (§3.4).

Augmentation. A Grain pipeline applies, per image, on worker threads: a horizontal flip with probability 0.5 (applied jointly to the image and its layout mask), then colour jitter with brightness shift $b \sim U(-0.2, 0.2)$, contrast $c \sim U(0.8, 1.2)$, saturation $s \sim U(0.8, 1.2)$, as $x \leftarrow x + b$; $x \leftarrow \bar{x} + c(x - \bar{x})$ with $\bar{x}$ the image mean; $x \leftarrow g + s(x - g)$ with g the per-pixel channel mean; then clipped to $[-1, 1]$. Masks pass through the jitter untouched. Batch size 128; batches straddle epoch boundaries (ceiling division), so an epoch is $\lceil N/128 \rceil$ batches.

Landmarks and masks. Each image is upsampled to 256×256 (bicubic) and passed to `hysts/anime-face-detector` (YOLOv3 face detector, HRNet 28-keypoint regressor) with the whole image as the face box. The 28 points are fixed in number and order: 0–4 chin contour left to right; 5–7 and 8–10 the two brows left to right; 11–16 and 17–22 the two eyes, each as the upper lid left to right (3 points) then the lower lid left to right (3 points); 23 nose; 24–27 mouth (left corner, top, right corner, bottom). Coordinates are stored in the image’s $[-1, 1]^2$ frame with the pixel-centre convention $((p + 0.5)/256) \cdot 2 - 1$; points the detector places outside the crop are kept (they occur for tight crops). Per-point confidences $\in [0, 1]$ are stored alongside (`experiments/extract_landmarks.py`).

The three-channel mask used for conditioning is rasterised from the landmarks: channel 0 the convex hull of points 0–10 (face), channel 1 the union of the hulls of 11–16 and

17–22 (eyes), channel 2 the hull of 24–27 (mouth). Hulls are filled at 256×256 and area-averaged to 64×64 , giving soft, anti-aliased edges; values are stored as `uint8` and used in `[0, 1]`. The eye channel covers $\approx 4.8\%$ of pixels. `data/layouts.py:rasterize` reproduces the stored masks with IoU 0.99 (face), 0.95 (eyes), 0.64 (mouth; a few-pixel region, quantisation).

Curation. Sorting images by mean landmark confidence and inspecting bands (`runs/ablation/curation_*`, below 0.3 the content is dominated by non-faces (torsos, clothing, hands, duplicated crops); the 0.3–0.65 band is legitimate hard faces (closed eyes, glasses, masks, mascots) that a generator should keep. Training therefore drops images with mean confidence < 0.3 (317 images, 1.47%; 21,234 remain, 166 batches per epoch). **FID reference statistics always use the full set**, so scores stay comparable across runs. Measured effect (§7): neutral to slightly positive.

3 Model

3.1 Flow matching with x-prediction

With $x_1 \sim p_{\text{data}}$, $x_0 \sim \mathcal{N}(0, I)$ and a time $t \in [0, 1]$, the interpolant is $x_t = t x_1 + (1 - t) x_0$ and the conditional velocity is $x_1 - x_0$. The network predicts the *clean image* $\hat{x} = f_\theta(x_t, t)$ (x-prediction), and the velocity is derived wherever it is needed as

$$u_\theta(x_t, t) = \frac{\hat{x} - x_t}{\max(1 - t, \epsilon_d)}, \quad \epsilon_d = 0.05. \quad (1)$$

The training loss is the plain conditional flow-matching loss in velocity space,

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, x_1} \frac{1}{2} \|u_\theta(x_t, t) - (x_1 - x_0)\|^2 = \mathbb{E} \frac{1}{2} \frac{\|\hat{x} - x_1\|^2}{\max(1 - t, \epsilon_d)^2}, \quad (2)$$

i.e. a per-pixel squared error on $\hat{x}$ with weight $w(t) = 1 / \max(1 - t, \epsilon_d)^2$ (≈ 1.2 at $t = 0.1$, 4 at 0.5, 100 at 0.9, 400 at the floor). The floor bounds the weight so that a single sample near $t = 1$ cannot own a batch’s gradient; this is the JiT recipe (Li & He, 2026). t is drawn logit-normal, $t = \sigma(\mu + \sigma_t z)$, $z \sim \mathcal{N}(0, 1)$, $\mu = -0.8$, $\sigma_t = 1$, which puts more mass at low t (high noise) and thins the tail at $t \rightarrow 1$ where $w(t)$ is large. The population minimiser of (2) at every (x_t, t) is the conditional mean $x^*(t) = \mathbb{E}[x_1 | x_t]$ (a fact used throughout §5).

The loss has no auxiliary terms. This was not the starting point; it is the conclusion of §5–7.

3.2 Conditioning token

For a conditioned model with $K = 3$ mask channels, the network input is the concatenation along channels

$$x_{\text{in}} = [x_t; m \odot \kappa; \kappa] \in \mathbb{R}^{H \times W \times (3 + K + 1)}, \quad (3)$$

where $m \in [0, 1]^{H \times W \times K}$ is the layout mask and $\kappa \in \{0, 1\}$ a per-sample indicator broadcast to a channel (`imagefm.cond_token`). The *null token* is $\kappa = 0$ with zero mask channels; the indicator disambiguates “unconditioned” from “a face with no parts”. Classifier-free

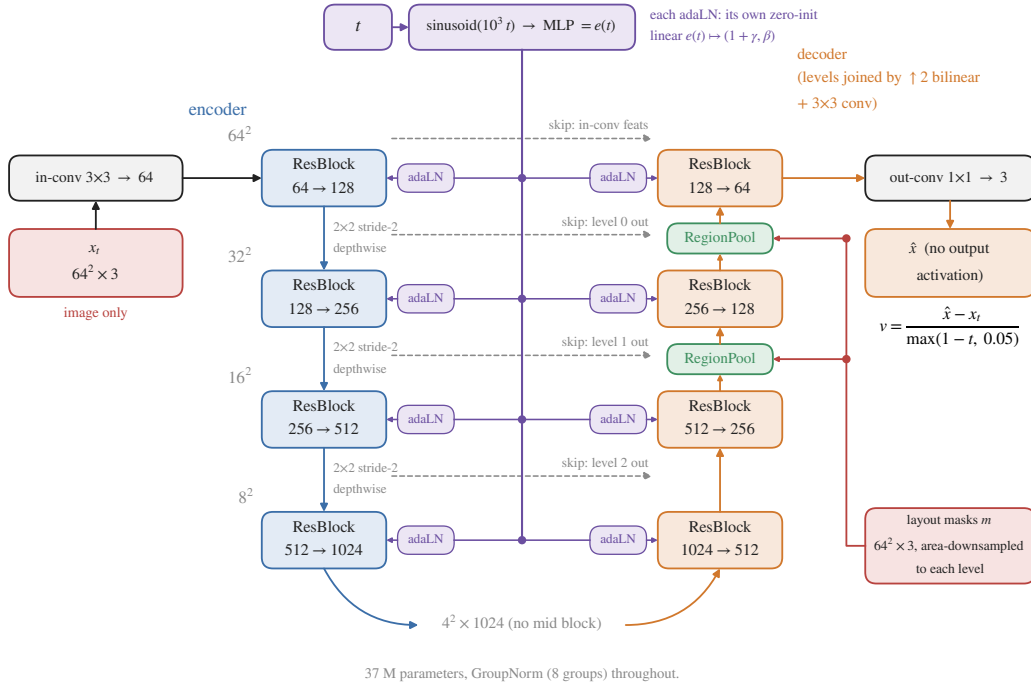


Figure 1: The velocity network. The image x_t is the only input to the U-Net; the layout masks m enter solely through the region-pooling layers after the decoder levels producing 16^2 and 32^2 . Four encoder levels (ResBlock doubling the channels, then a 2×2 stride-2 depthwise convolution), a bare $4^2 \times 1024$ tensor with no mid block, four decoder levels (bilinear $\uparrow 2$ and a 3×3 convolution, concatenation of the skip, ResBlock halving the channels). Skips run from the *input* of each encoder level to the decoder level after its upsample (Fig. 3). One shared time embedding $e(t)$ feeds a separate adaLN-Zero projection in every ResBlock; the output is $\hat{x}$ with no activation, and the velocity is derived from it.

training (Ho & Salimans, 2022) replaces the mask by the null token with probability p_{drop} (cond_dropout); the final recipe uses $p_{\text{drop}} = 0$ (a pure conditional model) because dropout cost ≈ 7 FID at the screening horizon (§12.1). The target is then $\mathbb{E}[x_1 \mid x_t, m]$: the layout is given and the network renders. Sampling never uses a real image’s mask; layouts come from the prior of §3.6.

3.3 Sampler

The probability-flow ODE $\dot{x} = u_\theta(x, t)$ is integrated from $t = 0$ to $t_1 = 1 - 10^{-3}$ with `diffraX`’s `Dopri5` on a fixed grid of n steps ($\Delta t = t_1/n$, no adaptive control), $n = 64$ for reported samples and $n = 16$ for FID during training and screening (§4). One solver (`imagefm._solve`) serves training-time samples, FID and figures. For a conditioned model the conditioning token is held fixed along the trajectory.

3.4 Network

The velocity network is a four-level U-Net (`models/unet.py`), channels-first internally, 36.7 M parameters at `base_channels=64` (9.38 M at 32).

- **Input/output.** A 3×3 convolution (padding 1) maps the $3 + K + 1 = 7$ input channels to 64; a 1×1 convolution maps the final 64 channels to the 3 output channels. There is **no final activation**: the velocity derived from $\hat{x}$ is unbounded regardless, and a bounded output would only add vanishing gradients.
- **Time embedding.** t is scaled by 10^3 and embedded with 64 frequencies ω_k geometrically spaced from 1 to 10^{-4} , $e(t) = [\sin(10^3 t \omega_k); \cos(10^3 t \omega_k)] \in \mathbb{R}^{128}$, then a two-layer MLP $128 \rightarrow 128 \rightarrow 128$ with SiLU. Without the 10^3 scale no frequency completes a cycle on $[0, 1]$ and the embedding is nearly constant in t .
- **ResBlock** (in c , out c'): GroupNorm(8 groups, $\epsilon = 10^{-5}$, statistics over the group's channels and space, learned affine) $\rightarrow$ SiLU $\rightarrow 3 \times 3$ conv $\rightarrow$ GroupNorm (no affine) $\rightarrow$ adaptive modulation $h \odot (1 + \gamma) + \beta$ with $[\gamma; \beta] = W_a \text{SiLU}(e) + b_a$, $W_a \in \mathbb{R}^{2c' \times 128}$ **zero-initialised** $\rightarrow$ SiLU $\rightarrow 3 \times 3$ conv, plus a skip path (1×1 conv if $c \neq c'$, identity otherwise). The zero initialisation follows adaLN-Zero (Peebles & Xie, 2023), the FiLM analogue of the zero- γ trick of Wu & He (2018); the modulation has no output gate. GroupNorm (Wu & He, 2018) rather than InstanceNorm is deliberate: per-channel statistics would strip per-image brightness and colour cast, which a generator must keep.
- **UBlock** at level i with channels $c_i = 64 \cdot 2^i$: *down* = ResBlock($c_i \rightarrow 2c_i$) $\rightarrow$ SiLU $\rightarrow$ depthwise 2×2 stride-2 conv; *up* = bilinear $\times 2 \rightarrow 3 \times 3$ conv ($2c_i \rightarrow c_i$) $\rightarrow$ SiLU $\rightarrow$ concatenate the skip (c_i channels) $\rightarrow$ ResBlock($2c_i \rightarrow c_i$). Resolutions 64, 32, 16, 8 at the four levels and a $4 \times 4 \times 1024$ bottleneck; the bottleneck is the last down-conv's output with no mid-block.
- **Skip layout (a documented deviation).** The skip saved at level i is the *input* to that level's down block, so level $i+1$'s skip is level i 's encoder output after the strided conv—the standard pattern shifted down one level: the top decoder level receives `in_conv` features rather than a ResBlock output, and each skip is half the standard width. Tested (§11): the standard layout is neutral at 9.4 M parameters and unstable at 37 M at the default learning rate; the deviation is kept.
- **Region pooling** (§3.5) after the up blocks that produce 16×16 (256 channels) and 32×32 (128 channels).

3.5 Mask-guided region pooling

Let $h \in \mathbb{R}^{C \times H \times W}$ be the decoder feature map after an up block and $m_k \in [0, 1]^{H \times W}$, $k = 1..4$, the layout masks area-averaged to $H \times W$ for the regions *face*, *eyes*, *mouth* and *hair/background* = $1 - \text{face}$ (read directly from the network's own input channels). The layer is

$$\bar{e}_k = \frac{\sum_p m_k(p) h(p)}{\sum_p m_k(p) + 10^{-6}} \in \mathbb{R}^C, \quad h' = h + \sum_{k=1}^4 m_k \otimes (W_k \bar{e}_k + b_k), \quad W_k \in \mathbb{R}^{C \times C}, b_k \in \mathbb{R}^C, \quad (4)$$

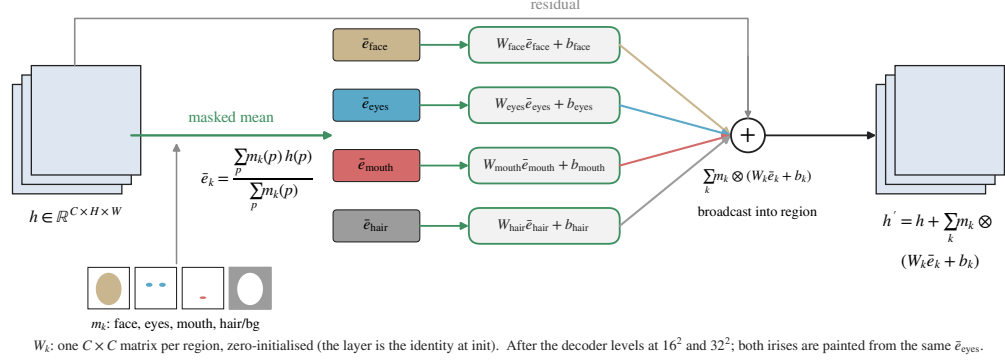


Figure 2: The region-pooling layer. Features are mean-pooled inside each soft region (face, eyes, mouth, hair/background), projected by a per-region zero-initialised $C \times C$ matrix and broadcast back into that region, as a residual. Both irises are painted from the same pooled eye vector.

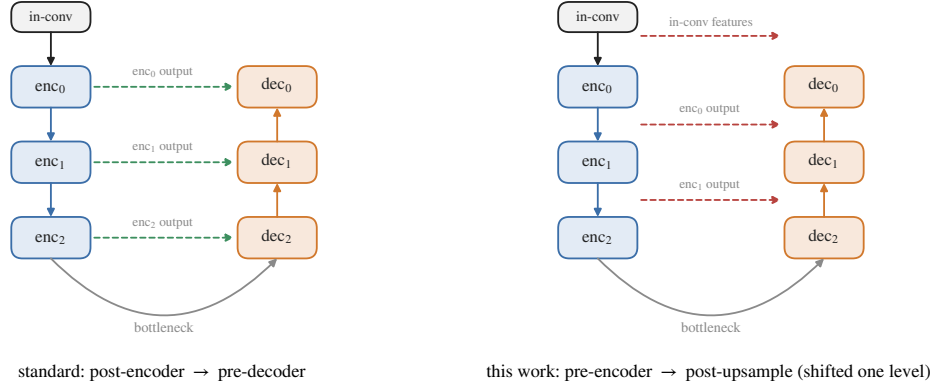
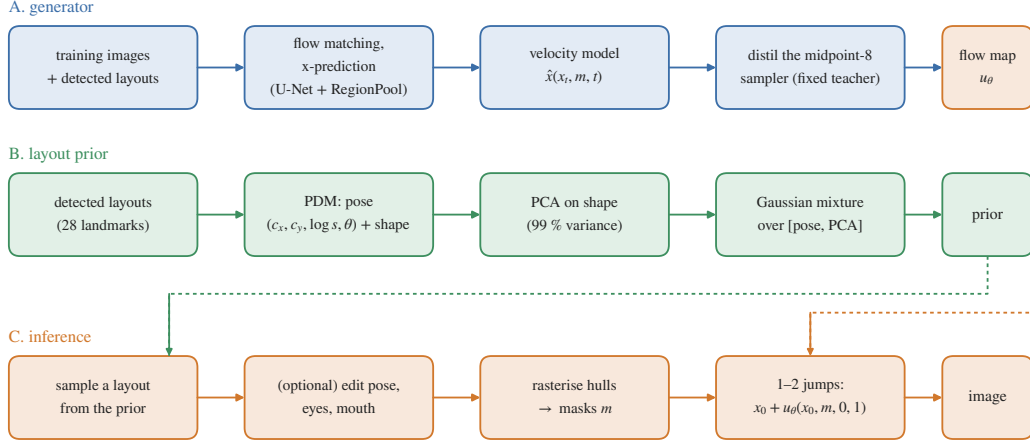


Figure 3: Skip wiring. Standard U-Nets pass each encoder level’s output to the matching decoder level; this network passes the level’s *input* (the previous level’s output, half the width), i.e. the standard pattern shifted down one level. Neutral at 9.4 M and the more stable choice at 37 M (§11).

with $W_k = 0$, $b_k = 0$ at initialisation (the layer is the identity at step 0), all regions pooled from the same input h (order independent), implemented as three einsums. Parameters: $4(C^2 + C)$ per level, 0.33 M in total at width 64.

Interpretation. In attention terms this is masked attention with a single fixed query per region and uniform weights: the *routing* (which positions belong together) is supplied by the layout, and only the map on the aggregate is learned. Both irises receive an identical additive term derived from both of them, so the two local renderers start from the same offset and—with zero initialisation—can only be moved together. The closest published relative is SEAN (Zhu et al., 2020), which broadcasts a per-region style code taken from an encoder of a reference image; here the code is pooled from the network’s own decoder features, enforcing internal consistency rather than external style. The layer acts at 16×16 and 32×32 because that is where iris colour is decided; a global vector at the 4×4 bottleneck and attention there were both tried and did not move the defect (§11).



A and B are fitted independently from the same detected layouts. Only C runs at inference time: no real image is involved.

Figure 4: The system. (A) The generator: flow matching with x-prediction on the training images and their detected layouts, then distillation of its midpoint-8 sampler into a one/two-jump flow map. (B) The layout prior, fitted independently from the same detected landmarks: a point-distribution model separates pose from shape, PCA keeps the 99% shape subspace, and a Gaussian mixture is fitted over pose and PCA coefficients. (C) Inference draws a layout from the prior (optionally edited), rasterises it to masks and renders in one or two network evaluations; no real image is involved.

History of the definition. The first implementation updated h sequentially over regions (later regions pooled features already offset by earlier ones; regions overlap, as the eyes lie inside the face hull). The parallel form (4) differs from it by a mean $|\Delta\hat{x}|$ of 0.008 (1.6% of $|\hat{x}|$) on a trained checkpoint; the positive result was re-established on the parallel form before the final run (§12.1).

3.6 Layout prior

Unconditional generation requires a layout that comes from no real image. The prior (`data/layouts.py`, fitted by `experiments/landmark_prior.py`) is a point-distribution model (Cootes et al., 1995) with a Gaussian-mixture density.

Parametrisation. For a landmark set $L \in \mathbb{R}^{28 \times 2}$: pose $(c_x, c_y, \log s, \vartheta)$ with c the centroid, s the inter-ocular distance (between the two eye centroids) and ϑ the angle of the eye line; shape = the points centred, rotated by $-\vartheta$ and divided by s , flattened to $\mathbb{R}^{56}$. The map is exactly invertible (round-trip error 2×10^{-7}).

Mirror augmentation. $x \rightarrow -x$ with the point permutation that reverses every horizontal run and swaps the left/right groups ([4, 3, 2, 1, 0, 10, 9, 8, 7, 6, 5, 19, 18, 17, 22, 21, 20, 13, 12, 11, 16, 15, 14, 23, 26, 25, 24, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56]). A wrong pairing shows as a large residual between the mean layout and the mean mirrored layout; the verified permutation gives a maximum per-point residual of 0.085 (the remainder is genuine dataset asymmetry: the image-left eye is systematically smaller, 0.356 vs 0.383, i.e. slightly turned faces).

Fit. On the 20,519 layouts with mean confidence ≥ 0.7 , doubled by mirroring: PCA on

Table 1: Layout prior: marginal statistics (mean / std, $[-1, 1]$ units) of real layouts, DP-GMM samples and a KDE baseline (real layouts + Gaussian jitter, $\sigma = 0.01$).

measure	real	DP-GMM	KDE
eye spacing	0.769 / 0.055	0.768 / 0.054	0.769 / 0.057
eye size A / B	0.356 / 0.383	0.367 / 0.369	0.369 / 0.369
openness A / B	0.604 / 0.564	0.583 / 0.582	0.587 / 0.588
openness A – B	0.137 / 0.131	0.131 / 0.122	0.148 / 0.140
mouth width	0.185 / 0.115	0.185 / 0.108	0.182 / 0.113
face width	1.424 / 0.092	1.422 / 0.091	1.422 / 0.092
off-canvas fraction	0.003	0.001	0.004

the shape vectors keeping 99% of variance (32 components), then a Dirichlet-process Gaussian mixture (variational, up to 16 components, full covariances, 500 iterations; `sklearn.BayesianGaussianMixture`) on [pose; PCA coefficients]; 16 components remain active. Sampling needs only numpy: choose a component by weight, draw from its Gaussian, invert the PCA and the pose normalisation, rasterise as in §2 (out-of-canvas points clipped only at rasterisation).

Checks (4,000 samples each; Table 1): the sampled marginals match the real ones, including the left/right openness *difference*, and a jitter-the-real-layouts baseline gives the same picture. Grids of sampled masks are plausible by eye, including the closed-eye mode.

3.7 Optimisation

Adam with global-norm gradient clipping at 1.0; batch 128; learning rate 10^{-3} peak with a linear warm-up over 500 steps (clamped to 10% of short runs) and a cosine decay to 10^{-5} at the last step (`optax.warmup_cosine_decay_schedule`). Weights published for sampling and evaluation are an exponential moving average with decay 0.999, warmed up as $\min(0.999, (1 + s)/(10 + s))$ at step s so early averages are not dominated by the initialisation (the scheme of the DDPM reference code, Ho et al., 2020; the decay is shorter than the literature’s 0.9999 because these runs are tens of thousands of steps, not hundreds of thousands). Seed 49 for every run; the dataset order, noise and t draws are all derived from it.

The schedule is a late addition: every earlier run used a *constant* 10^{-3} , which turned out to be at the edge of stability for the 37 M model. Three runs with an added module diverged in a single epoch after tens of epochs of steady descent (§11, §13); the plain model survived 200 epochs with a thin margin. Constant 3×10^{-4} gives a *lower* training loss but a *worse* FID at 30 epochs (63.9 vs 56.5), so a lower constant rate was not adopted; warm-up plus decay was.

4 Evaluation protocol

FID. Inception pool features (`fidax`), 5,000 generated images against statistics of the full 21,551-image set. Real-vs-real at 5,000 samples gives a floor of 2.1. **Noise:** two seeds of the same 30-epoch configuration scored 56.5 and 52.7; differences under ≈ 5 FID at short horizons are not evidence. **Sampler:** on the 200-epoch model, 16 / 32 / 64 Dopri5 steps

score 31.6 / 31.4 / 32.0; training-time and screening FID therefore use 16 steps at a quarter of the cost. Training batch 256 is *slower* per epoch than 128 (39.9 vs 29.0 s) on this GPU; 128 is kept.

Loss as the sensitive statistic. Across seeds the 30-epoch training loss varies by ± 0.0003 while FID varies by ± 4 ; for architecture changes that leave the objective unchanged, loss is the sharper screen.

Precision / recall. [Kynkäänniemi et al. \(2019\)](#), $k = 3$ nearest neighbours in Inception space, 5,000 vs 5,000; real-vs-real ceiling 0.774 / 0.765.

Iris-mismatch rate. Split the dataset-mean eye mask at the vertical midline into a left and a right region. For an image, take the mask-weighted mean chroma (C_b, C_r) (BT.601, from $[-1, 1]$ RGB) of each region and their Euclidean distance. A sample is “mismatched” if that distance exceeds the real data’s 95th percentile (0.1027), so real data scores $\approx 5\%$ by construction (two disjoint halves: 5.1% and 5.9%). Validated by inspection: the 32 most-flagged samples of the 200-epoch model are all genuine red/blue, yellow/blue, purple/green pairs in otherwise well-formed faces; the least flagged are matched. **This metric is uncorrelated with FID:** the best-FID unconditioned model has a 35.5% rate.

Local-detail proxies. Mean Sobel magnitude (“edge mass”) and mean ink map of samples relative to real images; and, for the bottleneck analysis, per-region prediction error and per-cell texture statistics (§10).

5 Auxiliary losses on $\hat{x}$: theory

The shipped model added to (2) an edge loss and (optionally) a line loss:

$$\mathcal{L}_{\text{sobel}} = \frac{1}{2} (\mathbb{E} |\nabla_x \hat{x} - \nabla_x x_1| + \mathbb{E} |\nabla_y \hat{x} - \nabla_y x_1|), \quad \text{Sobel gradients per channel, L1 (Mathieu et al., 2016),} \quad (5)$$

$$\mathcal{L}_{\text{line}} = \mathbb{E} |\text{ink}(\hat{x}) - \text{ink}(x_1)|, \quad \text{ink}(x) = \text{relu}(\text{softclose}_{3 \times 3}(\text{luma } x) - \text{luma } x), \beta = 10, \quad (6)$$

with weights 0.1 and 0 respectively.

Proposition 1 (Irreducibility). *Let $x^*(t) = \mathbb{E}[x_1 \mid x_t]$ be the minimiser of (2). For any $\mathcal{L}_{\text{aux}}(\hat{x}, x_1) \geq 0$ with $\mathcal{L}_{\text{aux}}(x_1, x_1) = 0$, the value $\mathcal{L}_{\text{aux}}(x^*(t), x_1)$ is in general strictly positive and cannot be reduced by any network without leaving the optimum of (2). Only the excess $\mathcal{L}_{\text{aux}}(\hat{x}) - \mathcal{L}_{\text{aux}}(x^*)$ is a usable signal.*

At low t , x^* is a blurry average; an edge or line loss against the sharp x_1 measures that blur, and its gradient at x^* points toward “sharper than the evidence supports”. Whether this is *bias* rather than harmless noise depends on the minimiser of the auxiliary term: L1 on gradient differences is minimised by the per-pixel *median* of the target gradient, which under uncertainty about edge position is nearer zero than the mean, so minimising an irreducible L1 penalty rewards *flatter* predictions (hedging). The line loss is L1 on a nonlinear map

Table 2: Discrimination: loss under a degradation, normalised by the loss against an unrelated image.

degradation	pixel L2	sobel	line
blur $\sigma = 1$	0.064	0.379	0.552
noise $\sigma = 0.05$	0.005	0.112	0.103
desaturate 50%	0.019	0.075	0.000
brightness +0.2	0.073	0.000	0.000
erase strokes	0.097	0.355	0.550
jitter 1 px	0.206	0.625	0.812

and is biased for that reason too. Section 6 estimates the reducible fraction empirically; §7 confirms the prediction in training.

Proposition 2 (Weighted losses). *For a per-pixel weight w and prediction c at fixed x_t ,*

$$J(c) = \mathbb{E}[w(c-Y)^2 \mid x_t], \quad \nabla J = 0 \Rightarrow c^* = \frac{\mathbb{E}[wY \mid x_t]}{\mathbb{E}[w \mid x_t]} = \mathbb{E}[Y \mid x_t] + \frac{\text{Cov}(w, Y \mid x_t)}{\mathbb{E}[w \mid x_t]}.$$

$c^* = \mathbb{E}[Y \mid x_t]$ iff w is conditionally independent of Y given x_t , i.e. a function of (t, x_t) only.

Proof sketch. Tower property, condition on the argument of the predictor (not on Y : conditioning on Y would “derive” $c = Y$, which no function of x_t can be); J is a strictly convex quadratic in c ; differentiate and expand $\mathbb{E}[wY] = \mathbb{E}[w]\mathbb{E}[Y] + \text{Cov}(w, Y)$. Equivalently, conditioning on the pair (x_0, x_1) : a weight $w(x_0, x_1)$ is the unweighted CFM loss under the re-weighted coupling $\tilde{p} \propto wp$, so by Lipman et al. (2023) (Thm. 2) the learned field transports noise to the data distribution *tilted* by w . The per-sample gradient factorisation $W \odot r$ is correct but does not make the original optimum stationary, because $\mathbb{E}[wr \mid x_t] = \text{Cov}(w, r \mid x_t) \neq 0$ for a stochastic coupling ($r = 0$ for every sample is unattainable). Weights of the form $\lambda(t)$ —the $1/(1-t)^2$ factor, any t -gate—are free; weights built from x_1 (Sobel magnitude, ink map, region masks) are not, unless the map is nearly constant across the posterior (true for the eye mask on aligned faces, false for edge maps at low t).

6 Offline tests (no training)

All on the 100-epoch 9.4 M checkpoint that was trained *with* the edge loss (a caveat noted below), 256 images, seed 0 (experiments/aux_signal.py on the archive branch).

Discrimination. $\mathcal{L}(x, d(x))/\mathcal{L}(x, x')$ for degradations d and a random other image x' (Table 2). Both losses are 5–6 $\times$ more sensitive than pixel L2 to stroke erasure (a 3×3 luma closing) and to a 1-px jitter, and blind to global colour: they measure what they claim.

Pixel-space descent. From $\text{blur}_1(x)$, 200 Adam steps on the pixels minimising $\text{L2} + w \mathcal{L}_{\text{aux}}$: L2 alone recovers ink mass by step 25 and 113 dB PSNR by step 200; every auxiliary term is slower (sobel 0.1: 60 dB; line 1.0: 47 dB). With a known target the terms carry nothing L2 lacks.

Table 3: Irreducibility estimate: auxiliary loss of $\hat{x}(t)$ versus that of the PSNR-matched blur $\text{blur}_{\sigma^*}(x_1)$, and the reducible fraction.

t	σ^*	sobel: $\hat{x} / \text{blur}$	line: $\hat{x} / \text{blur}$	reducible sobel / line
0.10	8.0	0.807 / 0.820	0.099 / 0.101	0 / 0
0.30	2.5	0.677 / 0.744	0.090 / 0.098	0 / 0
0.50	1.1	0.496 / 0.516	0.071 / 0.085	0 / 0
0.65	0.7	0.367 / 0.344	0.054 / 0.068	0.06 / 0
0.85	0.5	0.205 / 0.184	0.029 / 0.041	0.10 / 0

Table 4: Loss ablations, 9.4 M model, 40 epochs. “Class” is the verdict of Prop. 2.

arm	FID@39	ink/real	edge/real	class
no auxiliary (reference)	68.0	0.79	0.86	—
sobel L1, weight 0.1 (shipped default)	74.7	0.68	0.76	biased (median)
sobel L2, weight 0.05	68.0	—	—	unbiased (linear map) – inert
FM weight 1 + $ \text{Sobel}(x_1) /\text{mean}$	65.3	0.89	0.91	biased (tilt)
FM weight from $ \text{Sobel}(\text{sg } \hat{x}) $	65.3	0.79	0.87	unbiased
FM weight 1 + $\text{ink}(x_1)/\text{mean}$	97.0			biased; $\sim 50\times$ per pixel
FM weight 1 + m_{eye} , coef. 1.0 / 0.2	117.7 / 76.0			$\sim$ unbiased; $20\times / 5\times$

Irreducible fraction. For each t on a 0.05 grid, match the checkpoint’s $\hat{x}(t)$ with a Gaussian blur of x_1 at equal PSNR and read the blur’s auxiliary loss as an estimate of $\mathcal{L}_{\text{aux}}(x^*)$ (Table 3). The line loss of $\hat{x}$ is *below* the matched blur at every t ; the sobel loss is within $\pm 5\%$ except a 5–10% excess for $t \in [0.6, 0.85]$. At every t at least 90% of both penalties is unreachable, and no t -window contains signal worth gating for. With the default weights the auxiliary penalty is 0.6–1.1 $\times$ the velocity loss for $t < 0.5$. Caveats: a Gaussian blur is not $\mathbb{E}[x_1 \mid x_t]$ (it likely *over*-states the irreducible part), and the checkpoint was trained with the edge loss; hence the ablations of §7.

Gradient geometry. Cosine similarity of per-sample gradients with respect to $\hat{x}$: $\cos(\nabla \text{L2}, \nabla \text{sobel}) \approx 0.65$, $\cos(\nabla \text{L2}, \nabla \text{line}) \approx 0.40$ at every point tested. The sobel-L1 gradient is $\text{Sobel}^\top \text{sign}(\cdot)$: a saturated ± 1 stripe pattern over the whole image including flat background.

7 Training ablations of the losses

9.4 M model, seed 49, 40 epochs, FID every 10 unless stated.

Shipped default vs off (paired). Table 4: turning the edge loss *off* is better at every evaluation and the model trained *with* it generates *fewer* edges (edge mass 0.76 vs 0.86 of real; ink 0.68 vs 0.79). On training-time interpolants the two models agree within 1.5% ($\mathcal{L}_{\text{sobel}}(\hat{x}, x_1)$ at $t = 0.1$: 0.808 vs 0.821)—the sliver §6 called reducible—so the difference is how hedged predictions compound along the sampling trajectory. Decision: default 0.

Variants. The L2 Sobel term is exactly inert (68.0 vs 68.0), as predicted for an unbiased term with $\leq 10\%$ reducible signal. The two edge re-weightings end at the same FID whether

or not they carry the covariance term of Prop. 2; the tilt shows only in the proxies (edgier samples). Sparse-map weights (ink, eye) are harmful at every coefficient tried because the batch-mean normalisation turns “average extra weight 1” into $20\text{--}50\times$ on a few percent of pixels; the eye weight does reduce iris mismatch dose-dependently ($41.8 \rightarrow 37.5 \rightarrow 32.8\%$) at a prohibitive FID cost.

At length. 37 M model, 200 epochs, FID at 49/99/149/199: plain 43.1/36.3/36.5/32.0; with the unbiased edge re-weighting 42.9/37.7/35.9/33.4. The one loss-side candidate is a wash. A mask-prediction auxiliary head on the encoder (BCE on area-pooled masks, weights 0.25 and 1.0) was monotonically worse at 9.4 M over 100 epochs (57.6, 69.7 vs 53.5 best) and at 37 M (62.4 vs 56.5). **The pixel-space auxiliary family is closed** at both widths and all horizons tested.

8 Scaling and the noise floor

Width and length are the two levers that moved FID substantially: 9.4 M / 40 epochs 68.0; 9.4 M / 100 epochs 53.5; 37 M / 30 epochs 56.5; 37 M / 200 epochs 32.0, still falling ($36.5 \rightarrow 32.0$ over the last 50). A spectral check supports spending on width: the effective rank (entropy of the normalised singular values of each convolution reshaped to $(\text{out}, \text{in} \cdot k^2)$) is 0.88 of the maximum at 9.4 M and 0.85 at 37 M, i.e. the doubled width is used, not idle; from 30 to 200 epochs the stable rank rises ($13.8 \rightarrow 17.5$) while the entropy rank falls slightly ($0.853 \rightarrow 0.825$), the signature of training concentrating a few directions while spreading energy into the mid-spectrum. Curation (§2) gave 53.2 / loss 0.0875 against 56.5–52.7 / 0.0890 at 30 epochs.

9 Sampling-time edge guidance

Priors about *samples* belong at sampling time, where they cannot bias the learned field. With $E(\hat{x}) = \text{relu}(m_{\text{real}} - |\text{Sobel}(\hat{x})|)^2$ (the one-sided deficit against the data’s mean edge mass, $m_{\text{real}} = 1.30$),

$$g = \frac{\partial E(\hat{x}(x_t, t))}{\partial x_t}, \quad \hat{x}' = \hat{x} - \lambda \mathcal{K}[t_{\text{lo}} < t < t_{\text{hi}}] \frac{g}{\text{rms}(g)}, \quad v = \frac{\hat{x}' - x_t}{\max(1 - t, \epsilon_d)}, \quad (7)$$

gradient through the network, RMS-normalised per sample. Results (5,000-sample FID): 9.4 M model $67.4 \rightarrow 59.0$ at $\lambda = 0.02$ ($\lambda = 0.05/0.10/0.20$: 59.4/62.1/69.5, overshooting as edge mass passes real); 37 M 200-epoch model $31.4 \rightarrow 28.1$. Restricting guidance to a t -interval (Kynkäänniemi et al., 2024) helps slightly: on the 9.4 M model $\lambda = 0.05$ on $t \in (0.5, 1)$ gives 57.8, while $(0, 0.5)$ is the worst window (61.4). Guidance raises precision ($0.52 \rightarrow 0.58$) and leaves recall at 0.15: a mode-seeking correction, which bounds its value at a few FID. Cost: one backward pass per network evaluation. On the region-pool model of §13 (best_model.eqx, 34.6) the gain is the same: 32.5 at $\lambda = 0.02$ on the full window, 32.8–33.6 for the other (λ, window) pairs; precision $0.47 \rightarrow 0.49$, recall $0.125 \rightarrow 0.108$.

9.1 Autoguidance

Karras et al. (2024) guide a model with a worse version of itself,

$$v = v_{\text{good}} + w(v_{\text{good}} - v_{\text{bad}}), \quad (8)$$

where v_{bad} is a smaller or less-trained model of the same family: the extrapolation removes the errors the two models share and the bad one makes more strongly. For a layout-conditioned model both velocities see the same mask; the cost is one extra forward pass per evaluation and no training (`experiments/autoguide.py`). Good model: `wide_rp_300/best_model.eqx` (34.6 unguided), 5,000 samples, 16 steps, prior layouts.

bad model	window	$w=0.5$	0.75	1	1.25	1.5	2
<code>confirm_rp</code> (same recipe, 40 ep, FID 52)	(0, 1)	24.9	22.0	21.2	23.3	29.7	56.9
<code>confirm_rp</code>	(0.3, 1)	26.6		23.1			44.7
epoch 49 of the same run	(0, 1)	28.9		26.5			51.6

At the optimum ($w = 1$, full window): 64 sampler steps give the same 21.2; real layouts 20.8 (vs 34.0 unguided, so the prior-vs-real gap is unchanged); edge mass rises from 0.90 to 1.08 of real; on the best checkpoint of the one-segment run (§13, 33.2 unguided) the result is **18.1**. Three readings. (i) −13 FID is three to four times the gain of any energy guidance and moves the layout-conditioned model from 3.5 FID behind the schedule-matched unconditioned control to 11 ahead of it. (ii) The guide must be bad in the right way: a separately trained 40-epoch checkpoint of the same recipe beats epoch 49 of the *same* run—which shares the good model’s initialisation and early trajectory—by 5 FID at every w . (iii) The response is sharp above $w = 1$ (collapse at 2), and restricting the window, which helped energy guidance, hurts here. The rise in edge mass says what is being corrected: the hedged, low-contrast detail that §10 identifies as the model’s characteristic error, which an under-trained model makes more strongly. Combining with one refinement round (§13.2) *hurts* (25.2). Iris mismatch, precision/recall and mask-following under autoguidance have not yet been re-measured. The result reproduces on a second dataset (§14).

10 Where the model fails

On the 37 M 200-epoch unconditioned model (`experiments/bottleneck.py`).

Kind of gap. FID floor 2.1. Precision 0.521, recall **0.154** (ceiling 0.774/0.765). Coverage dominates: 85% of real images have no generated neighbour. The samples farthest from the real manifold are washed-out, low-contrast images with incoherent hair, clothing and background—averages of modes the model has not learned, not broken faces.

Where. Prediction error divided by per-pixel data variance (512 real pairs): at $t = 0.3$, face 0.41, eyes 0.43, **mouth** 0.50, hair/background 0.25; the mouth is recovered worst relative to what is predictable because small low-variance high-frequency features receive the least loss weight. Generated-vs-real local statistics on an 8×8 grid: edge mass −9 to −12%, luma

std -9 to -13% (worst at the mouth/nose centre, -30 to -40%), chroma std -16 to -21% everywhere. Colour *means* are right (skin chroma $-0.062/0.089$ vs $-0.070/0.098$); colour *spread* is 13% narrow and the chroma-histogram distance to real is $2.1\times$ the real-vs-real floor. A palette loss therefore has no target; what is missing is diversity, which no per-sample loss supplies.

Eyes. Iris mismatch 35.5% on this model (§4), i.e. a long-range colour-coupling failure that FID does not see and that motivated §12.

Region-pool model. The same analysis on `wide_rp_300/best_model.eqx` (§13): precision 0.471 , recall 0.125 —the same coverage picture, slightly worse on both axes at a similar FID. With the layout given, the variance-normalised error at $t = 0.3$ is even across regions (face 0.32 , eyes 0.30 , mouth 0.33 , hair 0.23 ; the mouth is no longer the outlier), and the local-statistics deficit is a uniform -9 to -12% on edge mass, luma and chroma std: the layout removes the *where* uncertainty and leaves the hedging.

11 Architecture variants that did not help

All paired, 37 M, 30 epochs unless stated; loss (stable to ± 0.0003) is the primary read-out.

- **Standard skips.** 9.4 M: 76.0 vs 76.0 (neutral). 37 M at constant 10^{-3} : FID 188 , loss flat at 0.108 from epoch 3, samples frozen into a tiled texture—an optimisation collapse; at 3×10^{-4} : 63.6 with loss $0.0874 < 0.0890$ but no better than the legacy layout. Kept legacy.
- **Mid-block (ResBlock–attention–ResBlock at 4×4).** 9.4 M, 40 epochs: 78.7 vs 77.1 . 37 M, 200 epochs, constant 10^{-3} : 75.8 with a loss spike to 0.635 at epoch 12—an optimisation failure, not a verdict on attention. It had the lowest iris mismatch of the unconditioned models (23.0%), consistent with global mixing helping the coupling.
- **Global code.** $g = W_2 \text{SiLU}(W_1 \text{LN}(\text{mean}_{hw} z) + b_1) + b_2$ from the bottleneck z , $W_2 = 0$ at init, added to the decoder’s time embedding (the ADM/DiT class-embedding recipe). Loss 0.0869 vs 0.0870 , FID 63.8 vs 63.9 , mismatch 43.8% vs 42.6% . The path *did* switch on ($\|g\| \approx 0.6\text{--}0.8 \|e\|$, image-dependent) and changed nothing: the decoder is not short of global *information*; the defect is in how two regions are *rendered*, which is what §3.5 addresses.

12 Layout conditioning and region pooling: results

12.1 Conditioning alone

37 M, curated. With dropout 0.15 (30 epochs): unconditional mode 62.6 , prior layouts 57.8 , real layouts 57.2 , mismatch 29.7% ; without dropout (40 epochs, loss 0.0813): prior 50.7 , real 50.1 , mismatch 33.6% . Unconditional references at 30 epochs: $56.5/52.7/53.2$. Layout information lowers the training loss but not FID (real and prior layouts agree, so the prior is not the limitation) and does not touch iris agreement: a layout says where the eyes are, not what colour they share. Dropout costs ≈ 7 FID at this horizon.

12.2 Region pooling

Same recipe (pure conditional, 40 epochs, LR 10^{-3} constant, seed 49, 16-step FID):

40 epochs	loss	FID prior / real layouts	iris mismatch prior / real
no region pool	0.0813	50.7 / 50.1	33.6% / 38.7%
region pool (sequential form)	0.0807	48.2 / 48.0	6.6% / 5.1%
region pool (parallel form, confirmation)	0.0815	52.1 / 51.4	6.25% / 5.1%
real data			5.0–5.9%

The mismatch rate falls to the data’s own rate. Paired samples (same noise, same layouts) are the same face pair-for-pair with only the iris pairs changed—matched and more saturated. The FID difference between the two region-pool runs (48.2 vs 52.1) is inside the single-seed noise. The eye-region chroma distribution of the region-pool model matches real data in spread (saturation std 0.067 vs 0.071; C_b , C_r std 85–95% of real) and in hue histogram (same dominant blue-green band, 32% in both), so the layer enforces *agreement*, not a particular colour.

Mask-following. Does the model draw the face where the layout says? 512 prior layouts are rendered by each model, the landmark detector is re-run on the samples, and the detected landmarks are scored against the layout the image was rendered from (`experiments/mask_following.py`; real images against their own landmarks are the ceiling).

model	landmark error all / eyes / nose / mouth	det. conf. nose / mouth	IoU face / e
conditional, no region pool (40 ep)	0.052 / 0.027 / 0.101 / 0.062	0.57 / 0.71	0.918 / 0.8
conditional + region pool (<code>best_model</code>)	0.046 / 0.026 / 0.091 / 0.053	0.64 / 0.75	0.931 / 0.8
real	0	0.82 / 0.85	1

Both models follow the layout closely (100% detection; the eye error of 0.027 in $[-1, 1]$ units is under one pixel at 64 px). Region pooling tightens nose and mouth placement by $\approx 10\%$ and raises the detector’s confidence in those features, but the remaining gap to real in nose/mouth legibility (0.64 vs 0.82) is the larger effect and is not a layout problem.

13 Final training run

Recipe (`just paper`): 37 M U-Net with region pooling, pure conditional on the three-channel layout, curated data, batch 128, EMA 0.999, Dopri5 16-step FID every 50 epochs with prior-sampled layouts, 300 epochs in total.

Course of the run. The first attempt ran at a constant 10^{-3} and diverged in a single epoch at epoch 80 (loss 0.0784 $\rightarrow$ 0.2837, samples destroyed, partial recovery to 0.107), after a steady descent to 0.0776 and FID 42.0 at epoch 49. The warm-up-plus-cosine schedule of §3.7 was introduced in response and the run resumed from the epoch-49 EMA weights with a fresh optimiser (Adam moments are not checkpointed). A second segment

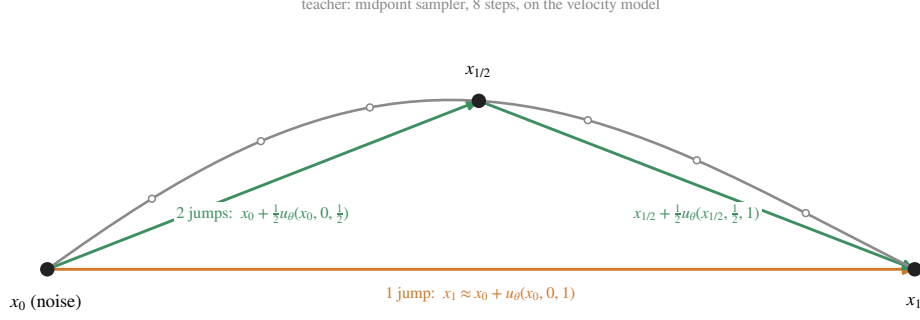
of 49 epochs (loss 0.0770) ended in a `cuSolver` failure inside the FID computation caused by a concurrent GPU job—an operational error, not a modelling one—and the run resumed again from those weights for the final 200 epochs. The effective schedule is three segments (49 constant, 49 and 200 warm-up-cosine), with the learning rate near its peak at both cut points, so it approximates one long cosine; checkpoints and loss logs of every segment are kept (`runs/wide_rp_300_ep49.eqx`, `_ep98.eqx`, and the corresponding `losses.json` files).

Results. Final loss 0.0557. Training-time FID over the last segment (16-step sampler, prior layouts), at cumulative epochs $\approx 147/197/247/297$: 47.8/34.9/36.2/37.5—best at ≈ 200 cumulative epochs, drifting up thereafter; `best_model.eqx` is that checkpoint. Final evaluation of the last weights (5,000 samples, 16 steps): FID **37.8** with prior layouts and 34.2 with real layouts; **iris mismatch** 4.7% under either (real data 5.0%; mean left/right chroma distance 0.039 vs 0.037 real). Layout adherence is strong (Fig. 6, layout-to-image panel: tilt, eye spacing and eye height track the mask in every sample). Against the plain unconditioned 200-epoch model (31.4 at the same sampler) the conditioned model is ≈ 6 FID behind at the end and ≈ 3.5 behind at its best point, while solving the eye defect outright; the prior-vs-real layout gap (3.5 FID), absent at 40 epochs (50.7 vs 50.1), opened as adherence sharpened, which points at a residual mismatch between the prior’s layouts and the data’s as one component of the deficit. Off-the-shelf anime super-resolution models (Real-ESRGAN variants, APISR) were tried on the samples for presentation and dropped: trained on clean-then-degraded pairs, they render the generator’s own errors as confident detail, and the native 64×64 samples read better.

The two confounds, resolved. Two further 200-epoch runs, each one segment of warm-up plus cosine with FID every 25 epochs: `ctrl_sched`, the plain unconditioned model on the same schedule (the missing control), and `wide_rp_200`, the region-pool recipe as recommended above.

epoch	24	49	74	99	124	149	174	199	final loss
plain, same schedule (<code>ctrl_sched</code>)	64.1	38.2	30.6	30.8	29.1	29.5	29.5	29.3	0.0789
conditional + region pool (<code>wide_rp_200</code>)	58.9	37.9	35.0	33.1	32.9	34.5	36.5	37.8	0.0537

`wide_rp_200/best_model.eqx` (epoch 124) evaluates at FID 32.9 with prior layouts, 31.4 with real layouts, iris mismatch 4.3% under either. Three conclusions. The schedule alone is worth ≈ 2.7 FID on the plain model ($32.0 \rightarrow 29.3$), whose curve is flat from epoch 75. The one-segment run reproduces the three-segment one (best 32.9 vs 34.9, final 37.8 vs 37.8): the warm restarts were not the cause of the drift. The drift after epoch ≈ 125 is real and specific to the conditioned model—its loss keeps falling ($0.065 \rightarrow 0.054$) while the control’s loss and FID are both flat—so the layout-conditioned region-pool model stands 3.5 FID behind the matched plain model at its best point and 8.5 behind at the end, at 4.3% iris mismatch against 35%. The drift is consistent with the model’s growing reliance on the region-pool broadcast (its contribution to $\hat{x}$ triples over training), which renders the prior’s imperfect layouts more literally: the prior-vs-real gap is 1.5 FID at the best checkpoint and 3.5 at the end. Autoguidance (§9.1) more than recovers the deficit: 18.1 on this checkpoint.



$$\mathcal{L} = E \|x_t + (s-t)u_\theta(x_t, m, t, s) - x_s^{\text{teacher}}\|^2, \quad (t, s) \in \{(0, 1), (0, \frac{1}{2}), (\frac{1}{2}, 1)\}$$

Figure 5: Fixed-teacher distillation. The velocity model under the midpoint sampler is a deterministic map from (noise, layout) to an image; the student – the same U-Net with a second time input – regresses its jumps $0 \rightarrow 1$, $0 \rightarrow \frac{1}{2}$ and $\frac{1}{2} \rightarrow 1$ onto the teacher’s states with a plain L2. Self-consistency objectives (MeanFlow, shortcut) either diverged or saturated when fine-tuned from this model (experiments/METHODS.md §3.4).

13.1 Distilled flow map: one or two evaluations

Sixteen Dopri5 steps cost ≈ 96 network evaluations; an explicit midpoint scheme with 8 steps (16 evaluations) matches it (FID 27.1 vs 29.3 on the plain model) and 4 steps (8 evaluations) gives 28.1, so below 8 evaluations a learned *flow map* $x_s = x_t + (s-t)u_\theta(x_t, t, s)$ is needed. Distilling the midpoint-8 sampler of the region-pool model into such a map (Fig. 5; 100,000 teacher trajectories, 35 epochs) gives FID 53.0 in one evaluation and **37.8** in two, against the teacher’s 32.9; exported to ONNX the two-jump sampler renders an image in 99 ms on two CPU threads, which is what the public demo runs.

13.2 Re-noise-and-re-solve refinement

A finished sample is re-noised to t_0 , $x_{t_0} = t_0 x + (1 - t_0) \epsilon$, and the ODE is solved again from t_0 with the same layout (SDEdit, Meng et al., 2022, on the model’s own output; one round of Restart sampling, Xu et al., 2023). On `best_model.eqx` with 16 steps per pass: $t_0 = 0/0.5/0.7/0.85$ gives FID 34.6/33.8/32.8/**31.8** with edge mass unchanged (0.89–0.90 of real). Unlike guidance it adds no contrast; it is a second draw of the late-decided detail. It does not stack with autoguidance ($21.2 \rightarrow 25.2$ with one round at $t_0 = 0.85$: the guided field is already off the model’s own manifold, and refinement pulls it back). It remains the in-distribution alternative to the external upscalers above.

14 Second dataset: CelebAMask-HQ

Does the recipe transfer to real faces? CelebAMask-HQ (Lee et al., 2020) provides 30,000 aligned CelebA-HQ faces with 19-class segmentation maps; we use the 256 px mirror, area-averaged to 64×64 , and build the four-channel layout from label unions: face (skin, nose,

wide_rp_300: 200 epochs, base_channels=64, cond_channels=3, region_pool=1, final FID 37.5



Figure 6: Showcase page produced by `generate_figures.py`: loss and FID curves, samples with layouts from the prior, layout-to-image pairs, real vs generated, noise interpolation at a fixed layout, and eye-region crops.

brows, eyes, lips, mouth, glasses), eyes, mouth (lips and interior), nose; region pooling adds 1 – face as before (`data/celebamask.py`). Curation drops 1.8% (duplicate thumbnails, empty eye/mouth/nose masks, mis-aligned or greyscale images), leaving 26,487. No layout prior is fitted: sampling-time layouts are the real label maps of a held-out bank (the last 3,000 ids, never trained on), so every number below is in the “real layouts” mode; FID statistics are the full 30,000 images. At 64 px the two eyes together cover one 16×16 cell ($14\times$ smaller than anime eyes), so the iris mechanism is barely testable at this size; a 128 px run (arrays built, fits the GPU at a 0.95 memory fraction) was not pursued, being a different cost class for a question the anime result already answers.

Training. Same architecture (37 M, `cond_channels` 4, `region_pool` 1), batch 128. At a constant 10^{-3} the run diverged at epoch 25 (loss $0.043 \rightarrow 0.285$, FID $41 \rightarrow 228$),

the same failure as the anime run and earlier; with a 5×10^{-4} peak, a 2,000-step warm-up and cosine decay, 120 epochs reach FID 39.9/27.3/27.3 at epochs 39/79/119 and a final loss of 0.036 (just celeba). The FID curve is flat from epoch 80: with real layouts there is no prior mismatch to render, and no late drift.

Autoguidance. Good model: the 120-epoch checkpoint (27.4 unguided); bad model: the epoch-24 checkpoint of the diverged constant-LR run (FID 41). FID at $w = 0.8/0.9/1.0$: 18.6/19.1/20.3, edge mass 1.07/1.10/1.13 of real (0.96 unguided). The anime result (§9.1) reproduces with a different mask source and a different guide: -9 FID, optimum near $w = 1$, edge mass crossing real at the optimum.

15 Decisions and their evidence (summary)

- **x-prediction, velocity-space loss, floor 0.05, logit-normal t :** JiT recipe; the parametrisation is what makes losses on $\hat{x}$ possible, and §5 is why none are used.
- **No auxiliary losses:** Prop. 1–2, Tables 3–4, the 200-epoch wash. Closed at both widths.
- **Guidance instead:** edge guidance -2 to -8 FID at $\lambda \approx 0.02$; refinement -3 at $t_0 = 0.85$; **autoguidance** -13 with a 40-epoch checkpoint of the same recipe at $w = 1$ —the largest single gain measured, and reproduced on CelebAMask-HQ (-9). All reported separately from unguided numbers.
- **GroupNorm (8 groups), zero-init adaLN, no output gate, no final activation, time scale 10^3 :** documented in code; the zero-init adaLN is why every added module (region pool, global code) can be identity at initialisation.
- **Legacy skips kept:** neutral at 9.4 M, unstable at 37 M.
- **Width 64, batch 128, 16-step evaluation, curation on:** §8, §4.
- **Warm-up + cosine:** four single-epoch divergences at constant 10^{-3} (three anime, one CelebA); worth 2.7 FID on the plain model by itself.
- **Pure conditional (no dropout):** dropout ≈ 7 FID at the screening horizon; the prior makes the unconditional mode unnecessary.
- **Region pooling at 16×16 and 32×32 , four regions, zero-init, parallel form:** §12; the only structural change with a reproduced, metric-validated win (iris mismatch $35\% \rightarrow 4.3\text{--}4.7\%$, mask-following up), at 3.5 FID against the schedule-matched control before guidance and 11 FID ahead of it after autoguidance.
- **Layout prior: PDM + PCA(99%) + DP-GMM:** Table 1; sampling is numpy-only.

Open. (1) Autoguidance is under-explored: which guide (epoch, width, data fraction) is best—a separately trained 40-epoch checkpoint beats an early checkpoint of the same run by 5 FID; whether the iris rate, mask-following and recall survive $w = 1$; and whether the plain model gains as much (if not, conditioning, region pooling and autoguidance are one recipe; if so, autoguidance is orthogonal). (2) The conditioned model’s late-training drift: a better layout prior (real landmarks with jitter) or stopping at the FID-optimal checkpoint. (3) Coverage (recall 0.15) remains the largest gap of the unguided model and moves with

steps and capacity; EDM-style augmentation conditioning with hair/eye hue rotation (a dataset-specific coverage prior, semantically valid only for this domain), spatial mixing at 16×16 with warm-up, a hair-colour consistency metric for the region-pool layer, and flip-equivariant sampling. Shelved with reasons recorded: structured output heads (line/fill, palette) as precision-side bets with a representational risk.

Reproducibility

Repository: `tinyflow`, branch `feat/pixel-unet`; the complete working tree of the campaign, including every dropped variant and the offline harness, is preserved on branch `exp/aux-loss-campaign`. `experiments/METHODS.md` holds every table of this paper with the commands that produced it; just `paper NAME 300 50` reproduces the final run and the figures. Hardware: one NVIDIA RTX 4090 (24 GB); training throughput ≈ 22 s per epoch (166 batches of 128) for the 37 M model; a 16-step 5,000-sample FID takes ≈ 2 minutes.

References

- T. F. Cootes, C. J. Taylor, D. H. Cooper, J. Graham. Active shape models—their training and application. *CVIU*, 1995.
- J. Ho, A. Jain, P. Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 2020.
- J. Ho, T. Salimans. Classifier-free diffusion guidance. *arXiv:2207.12598*, 2022.
- T. Karras, M. Aittala, T. Kynkäänniemi, J. Lehtinen, T. Aila, S. Laine. Guiding a diffusion model with a bad version of itself. *NeurIPS*, 2024.
- T. Kynkäänniemi, T. Karras, S. Laine, J. Lehtinen, T. Aila. Improved precision and recall metric for assessing generative models. *NeurIPS*, 2019.
- T. Kynkäänniemi et al. Applying guidance in a limited interval improves sample and distribution quality in diffusion models. *NeurIPS*, 2024.
- C.-H. Lee, Z. Liu, L. Wu, P. Luo. MaskGAN: towards diverse and interactive facial image manipulation. *CVPR*, 2020.
- T. Li, K. He. Back to basics: let denoising generative models denoise (JiT). *CVPR*, 2026.
- Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le. Flow matching for generative modeling. *ICLR*, 2023.
- C. Meng, Y. He, Y. Song, J. Song, J. Wu, J.-Y. Zhu, S. Ermon. SDEdit: guided image synthesis and editing with stochastic differential equations. *ICLR*, 2022.
- M. Mathieu, C. Couprie, Y. LeCun. Deep multi-scale video prediction beyond mean square error. *ICLR*, 2016.
- W. Peebles, S. Xie. Scalable diffusion models with transformers. *ICCV*, 2023.
- Y. Wu, K. He. Group normalization. *ECCV*, 2018.
- Y. Xu, M. Deng, X. Cheng, Y. Tian, Z. Liu, T. Jaakkola. Restart sampling for improving generative processes. *NeurIPS*, 2023.

P. Zhu, R. Abdal, Y. Qin, P. Wonka. SEAN: image synthesis with semantic region-adaptive normalization. *CVPR*, 2020.